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# Classical and quantum computing

The fundamental limitations of any form of computation can be expressed in terms of the resource requirements of standard computational tasks under it. Within traditional models of computation, such as the Turing machine model, many problems are found to be intractable due to the limited computational capabilities of classical physical systems. However, quantum systems allow the range of tractable computations to be extended beyond that achievable by classical computation because the superposition principle offers a radically different sort of computational parallelism. The *quantum circuit model* (or *gate array model*), in which networks composed of quantum logic gates act on sets of qubits, is the dominant model of quantum computation and has an equivalent (quantum) Turing machine model. Both of these models and the general principles of quantum computation are discussed in this chapter. A number of specific algorithms, which illustrate the novel character of quantum computation, are described in the following chapter.

Any good model of computation will allow arbitrary operations to be accurately performed through a set of basic operations, as the quantum circuit model does. A very simple picture of how quantum computation works in a qubit architecture, in which operations are performed on elements of a Hilbert space with a dimension that is a power of 2, is the following. One begins with classical input information, that is, bits encoded in qubits in the computational basis, and a number of ancillary qubits each generally assumed to be in the standard state $|0\rangle$. In the quantum circuit model, a sequence of logic operations described by unitary transformations then takes place on this full collection of qubits, in parallel, as a result of which their values may be changed. After some specific number of logical operations, one measures designated output qubits to obtain the result (*readout*) of computation.

In this chapter, we begin by discussing Turing machines of various sorts, quantum as well as classical. We then examine computational complexity in quantum computing. Finally, we briefly consider fault tolerance in quantum computing and a specific proposal for accomplishing quantum computation in linear optics.

## 13.1 Classical computing and computational complexity

Alan Turing was responsible for introducing what is now known as the *Turing machine*, an abstract machine he considered to be a universal computing device in the sense that it is assumed to be capable of simulating any other conceivable computing device. Specifically, he assumed that the class of functions that can be evaluated via algorithms is identical to the class of computable functions, an assumption that has been taken both as a fundamental principle and an empirical hypothesis and is now known as the *Church–Turing thesis* [107, 426].

The Turing machine is now understood to be universal in at least two respects. First, it provides a universal definition of computability in the sense of computability in Lambda calculus.[1] Secondly, all reasonable computing machines—by which is meant all machines that can in principle be physically realized—are polynomially equivalent to Turing machines: the minimal time required for any reasonable machine and for a Turing machine to produce an output given an input of a given size are polynomially related. That such a *single* universal device could actually exist still remains open to question. It is noteworthy, for example, that the running times of Turing machines are not predictive of those of digital computers that use floating point numbers to address scientific problems, as is commonly the case in practice. In particular, such computations have computational costs associated with operations that are *independent of* operand size, unlike Turing machines. Nonetheless, the Turing machine model *is* a central element of the dogma of current computer science and, as such, is a good model in which to consider quantum computation here. Furthermore, Turing machines have the very attractive feature of capturing the intuitive essence of computation in a simple schema.

Because quantum computing devices have other properties not likely foreseen by Turing, they provide a ground for scrutinizing the Church–Turing thesis. For example, the Turing machine is a finite machine not designed to address continuous mathematical models. Richard Feynman touched on the question of the adequacy of the Turing machine model from the physical point of view when he raised the question of the ability of quantum systems to perform efficient simulations of other physical systems, particularly *quantum systems* [170].

The question of the difficulty of performing a computational task, such as computing the value of a function for a given argument, can be addressed both qualitatively and quantitatively. A function is *computable* if there exists an algorithm for computing it. Computational complexity theory seeks *bounds* on the resources necessary for solving computational problems.[2] The *computational complexity* of obtaining the value of a function can be given in terms of the number of computational steps required to calculate it for a

[1] For information regarding Lambda calculus, see [25].
[2] It has, as a result, been referred to as the *thermodynamics of computation* [318].